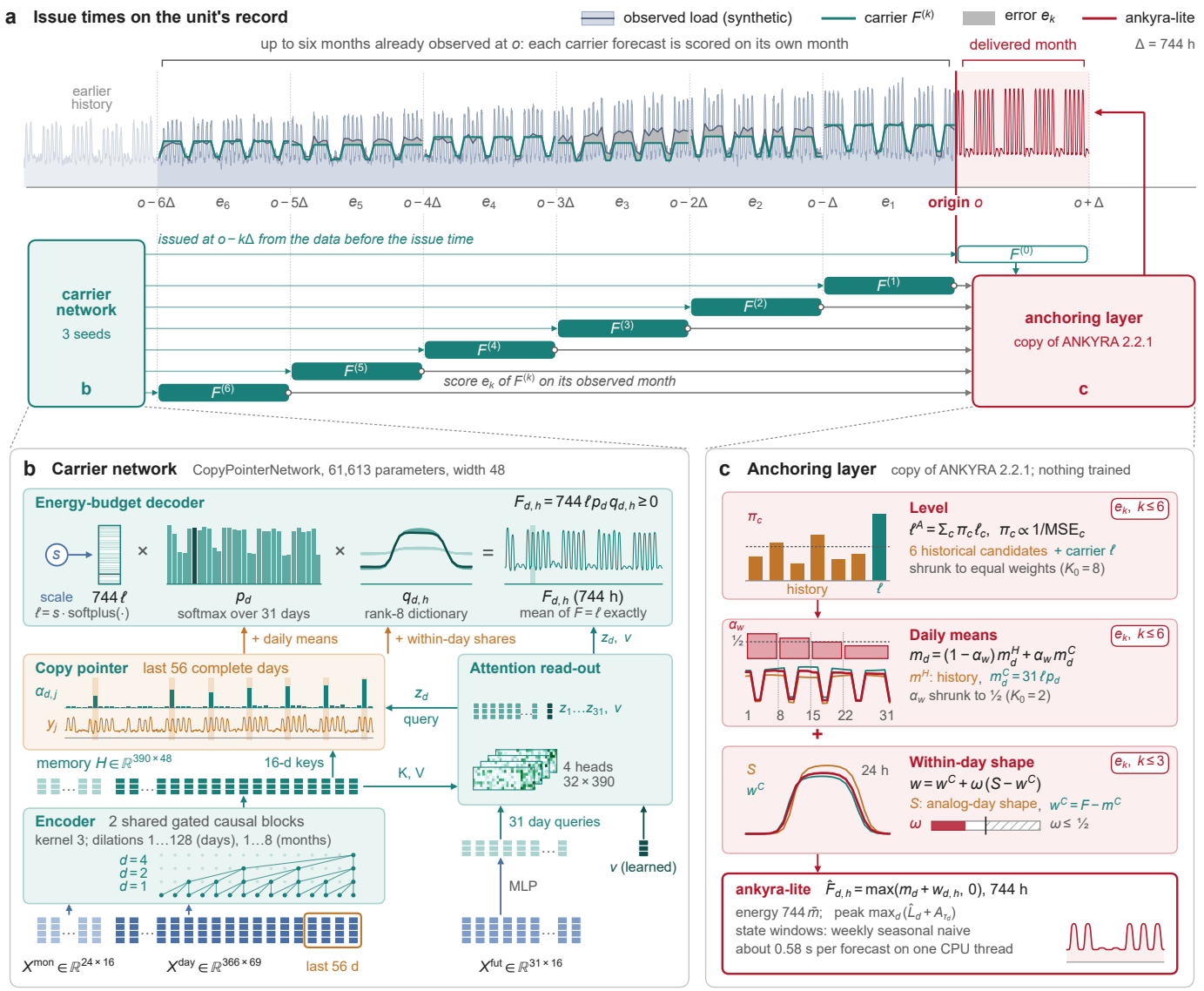

a Issue times on the unit's record
observed load (synthetic)
carrier $F^{(k)}$
error e_k
ankyra-lite
up to six months already observed at o: each carrier forecast is scored on its own month
delivered month
Δ = 744 h
earlier history
o − 6Δ
e6
o − 5Δ
e5
o − 4Δ
e4
o − 3Δ
e3
o − 2Δ
e2
o − Δ
e1
origin o
o + Δ
issued at o − kΔ from the data before the issue time
carrier network
3 seeds
b
F(0)
F(1)
F(2)
F(3)
F(4)
F(5)
F(6)
score e_k of $F^{(k)}$ on its observed month
anchoring layer
copy of ANKYRA 2.2.1
c
b Carrier network
CopyPointerNetwork, 61,613 parameters, width 48
Energy-budget decoder
$F_{d,h} = 744\,\ell\, p_d\, q_{d,h} \geq 0$
scale
744 ℓ
ℓ = s · softplus(·)
p_d
softmax over 31 days
$q_{d,h}$
rank-8 dictionary
$F_{d,h}$ (744 h)
mean of F = ℓ exactly
+ daily means
+ within-day shares
z_d, v
Copy pointer
last 56 complete days
$\alpha_{d,j}$
y_j
z_d
query
Attention read-out
$z_1 \ldots z_{31}$, v
memory $H \in \mathbb{R}^{390\times 48}$
16-d keys
K, V
4 heads
32 × 390
Encoder
2 shared gated causal blocks
kernel 3; dilations 1…128 (days), 1…8 (months)
d = 4
d = 2
d = 1
31 day queries
v (learned)
MLP
$X^{mon} \in \mathbb{R}^{24\times 16}$
$X^{day} \in \mathbb{R}^{366\times 69}$
last 56 d
$X^{fut} \in \mathbb{R}^{31\times 16}$
c Anchoring layer
copy of ANKYRA 2.2.1; nothing trained
π_c
history
ℓ
Level
e_k, k ≤ 6
$\ell^A = \Sigma_c\, \pi_c \ell_c$, $\pi_c \propto 1/\mathrm{MSE}_c$
6 historical candidates + carrier ℓ
shrunk to equal weights (K_0 = 8)
α_w
½
1
8
15
22
31
Daily means
e_k, k ≤ 6
$m_d = (1-\alpha_w)\, m_d^H + \alpha_w\, m_d^C$
m^H: history, $m_d^C = 31\,\ell\, p_d$
α_w shrunk to ½ (K_0 = 2)
S
w^C
24 h
Within-day shape
e_k, k ≤ 3
$w = w^C + \omega\,(S - w^C)$
S: analog-day shape, $w^C = F - m^C$
ω
ω ≤ ½
ankyra-lite
$\hat{F}_{d,h} = \max(m_d + w_{d,h},\, 0)$, 744 h
energy 744 $\tilde{m}$; peak $\max_d(\hat{L}_d + A_{\tau_d})$
state windows: weekly seasonal naive
about 0.58 s per forecast on one CPU thread